Tezpur University: Spring Semester 2018: Formal Languages & Automata (CO205)
Test: I :: Marks: 25 :: Time: 45 mins

1. Fill in the blanks [5]

   (a) A string is a __________ chosen from some alphabet.

   (b) ε is the identity for __________ operation in formal languages.

   (c) A language accepted by a finite automaton is called a __________.

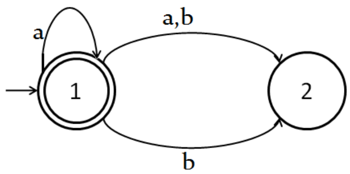
   (d) True/false: *In a DFA, it is possible to make a transition without seeing any input symbol*: __________.

   (e) True/false: *There can be more than one start state in an NFA*: __________

2. Given a string w, state the acceptance criteria for a DFA and an NFA. [2+2]

3. Construct a Deterministic Finite Automata $A = (Q, \Sigma, \delta, q_0, F)$ such that $L(A) = \{x | x \text{ is a binary number and is divisible by } 4\}$.
   Draw the transition diagram. [8]

4. Construct an equivalent DFA for the following NFA. [8]



Tezpur University: Spring Semester 2018: Formal Languages & Automata (CO205)
Test: I :: Marks: 25 :: Time: 45 mins

1. Fill in the blanks [5]

   (a) A string is a __________ chosen from some alphabet.

   (b) ε is the identity for __________ operation in formal languages.

   (c) A language accepted by a finite automaton is called a __________.

   (d) True/false: *In a DFA, it is possible to make a transition without seeing any input symbol*: __________.

   (e) True/false: *There can be more than one start state in an NFA*: __________

2. Given a string w, state the acceptance criteria for a DFA and an NFA. [2+2]

3. Construct a Deterministic Finite Automata $A = (Q, \Sigma, \delta, q_0, F)$ such that $L(A) = \{x | x \text{ is a binary number and is divisible by } 4\}$.
   Draw the transition diagram. [8]

4. Construct an equivalent DFA for the following NFA. [8]